

RESEARCH ARTICLE

Bio-Inspired High-Low Toughness Multilayer Mechanoluminescent Composite With Superior Mechanical Properties

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ABSTRACT

Polydimethylsiloxane confers stretchability, enhanced brightness, mechanical responsivity and self-charging capability to mechanoluminescence materials, yet its intrinsic low toughness and crack resistance pose a major obstacle to the practical application of ML devices. Herein, high-toughness silicone-rubber layers were introduced to slice the ZnS:Cu@Al₂O₃/PDMS matrix into a multilayer architecture that mimics the hinged microstructure of clam. The modified composite exhibits a dramatic enhancement in stretchability, with fracture strain increasing from approximately 100% to 500%. This improvement is accompanied by more than a four-fold boost in ultimate ML intensity, along with sustained durability demonstrated over 30 000 cycles at the tensile limit. The reduced layer thickness and suppressed crack propagation by the high-toughness silicone-rubber interlayer raise toughness from 0.95 to 4.93 MJ/m³. This work provides an effective strategy for enhancing the performance of PDMS-based flexible ML composites, advancing their practicality and offering new insights for applications in stress visualization, intelligent information display, and mechanics sensing.

1 | Introduction

Mechanoluminescence (ML), the phenomenon of light emission induced by mechanical stimuli, has garnered significant attention for its potential in self-powered sensing, stress visualization, and interactive displays [1–9]. To further enhance their utility, flexible ML devices capable of bending, twisting, and stretching are commonly realized by embedding inorganic ML phosphors into elastic polymer matrices, which provide the necessary biocom-

patibility and ease of processing for practical scenarios [10–16]. Among these matrices, polydimethylsiloxane (PDMS) serves as an essential and nearly irreplaceable matrix for mechanoluminescent composites, owing to its excellent optical transparency and ease of molding, as well as its ability to activate the ML through effective mechanical interaction with phosphors via interfacial friction and peeling [17–22]. However, the inherent mechanical limitations of PDMS, such as poor toughness and crack resistance pose a significant challenge in practical applications [23–26].

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The ZnS:Cu@Al₂O₃/PDMS system has emerged as a benchmark model in flexible ML composites, sustaining efficient luminescence over tens of thousands of stretching cycles until fracture occurs [27–30]. In this composite, ZnS:Cu@Al₂O₃ serves as a representative alternating current electroluminescent (ACEL) material and is capable of responding to weak electric fields or charges, a characteristic widely employed in electrostatic detection applications [31–34]. Through frictional contact with Al₂O₃, the PDMS matrix leverages its superior triboelectric charge accumulation capability to significantly amplify this ACEL effect, yielding bright ML. However, the performance ceiling of the system is intrinsically limited by the mechanical failure of the PDMS matrix, where cracks initiate and propagate readily under cyclic loading, ultimately restricting the maximum achievable strain and ML output. The inherent mechanical weakness of PDMS primarily stems from its linear polymer architecture and the relatively weak intermolecular interactions between its flexible siloxane backbones [35, 36]. Silicone rubber (SR) is a high-toughness derivative of PDMS which overcomes these limitations through a tailored 3D cross-linked network and reinforcement strategies ranging from interpenetrating polymer networks (IPNs) to the incorporation of dynamic bonds [37–40]. Nevertheless, its composites exhibit significantly lower luminescence performance compared to PDMS-based systems, presenting a critical trade-off between mechanical robustness and optical efficiency [41–43].

Learning from nature has been considered as one of the most effective strategies for improving material properties and functions [44]. Many force-bearing natural biological materials exhibit excellent mechanical properties, which are not derived from the constituent materials themselves, but rather attributed to their ordered structures [45–48]. For instance, bones exhibit both high strength and toughness due to the multiple mechanisms at its seven levels of structural hierarchy [49]. Similarly, spider silk boasts high tensile strength and large extensibility, thanks to its elaborate hierarchical microstructure [50]. Additionally, the hierarchical design of the hinge endows the shell with the ability to endure hundreds of thousands of repeated opening-and-closing motions over their lifetime [51]. The staggered arrangement of hard and soft materials allows hard components to withstand tensile loads, while soft materials mediate load transfer through shear deformation. Such multilayer structure leverages the inherent properties of each component, offering a promising approach for developing flexible ML film with excellent mechanical properties and crack resistance [21, 52–55].

Inspired by the above considerations, in this work, we alternate ZnS:Cu@Al₂O₃/PDMS functional layers with high-toughness SR interlayers to form a bioinspired multilayer structure to mimic the hinge of clams. The SR layers not only enhance interfacial bonding via their chemical similarity to PDMS but also effectively suppress crack propagation and delamination. As a result, the multilayer composite exhibits a remarkable increase in fracture strain (from ~100% to 500%), toughness (from 0.95 to 4.93 MJ m⁻³), and ultimate ML intensity (by over four times), along with excellent durability exceeding 30 000 cycles at large strains. This study provides an effective route to design high-performance flexible ML composites with integrated mechanical robustness and luminescent efficiency, paving the way for practical applications

in wearable devices, structural health monitoring, and secure anticounterfeiting.

2 | Results and Discussion

2.1 | Demonstration of the Multilayer Structure

Figure 1A illustrates the typical hinge tissue structure of bivalve shells. The bivalve shell achieves remarkable deformability and exceptional fatigue resistance because of its unique concentric laminated architecture, which comprises a highly ductile organic matrix reinforced with hard nanowires, enabling it to endure hundreds of thousands of repetitive opening-closing cycles throughout its lifespan. To mimic this multilevel composite architecture, high-toughness silicone rubbers were intentionally engineered to separate the original ZnS:Cu@Al₂O₃/PDMS functional layers, thereby establishing a series of multilayer heterogeneous structures. Figure 1B shows the representative image of the nonuple-layer (9L) engineered sample, in which the alternating layers are distinctly visible, demonstrating the uniform thickness and robust interfacial bonding while retaining exceptional flexibility. The micrographs of the engineered samples with varying layers under daylight and UV radiation are illustrated in Figure 1C. Smooth interfaces and strong layer-to-layer bonding can be clearly observed in all multilayer films, indicating that a high-quality laminated structure has been successfully achieved for all samples. Notably, to prevent alterations in mechanical properties that may arise from variations in thickness, the total thickness of different samples was uniformly fixed at approximately 1 mm, an optimal value as illustrated in Figure S1. Constrained by the overall thickness limitation, the 9L structure is the currently achievable processing limit, and further subdivision is not feasible due to the particle size limitation of the ZnS:Cu@Al₂O₃ phosphor in the functional layer (Figure S2). The multilayer system perfectly preserves the pristine structure's mechanical response, including shear/abrasion resistance, without any attenuation from the silicone top layer (Figure 1D and Movie S1). In addition, remarkable enhancement of stretchability and ML properties was realized through the multilayer design, as shown in Figure 1E and Movie S2. A significant elongation exceeding 400% was attained by the 9L composite film, representing more than a four-fold enhancement in ultimate brightness relative to the single-layer film of ZnS:Cu@Al₂O₃/PDMS (~100% elongation). Furthermore, when the membrane elongation exceeds 100% strain, a pronounced enhancement in ML intensity becomes visually detectable. This phenomenon conclusively demonstrates that the inferior fracture toughness critically constrains the maximum achievable ML in the ZnS:Cu@Al₂O₃/PDMS system.

2.2 | Mechanical Performance

The structure-dependent stretchability of ML composites was systematically evaluated. As illustrated in Figure 2A, stress-strain curves were collected for the ML strips from the single-layer (ZnS:Cu@Al₂O₃/PDMS) to the 3L, 5L, 7L, and 9L samples. The ultimate tensile strain improved sharply from ~130% to 500%, as the ZnS:Cu@Al₂O₃/PDMS functional layers were progressively subdivided into discrete microcompartments by the SR. In

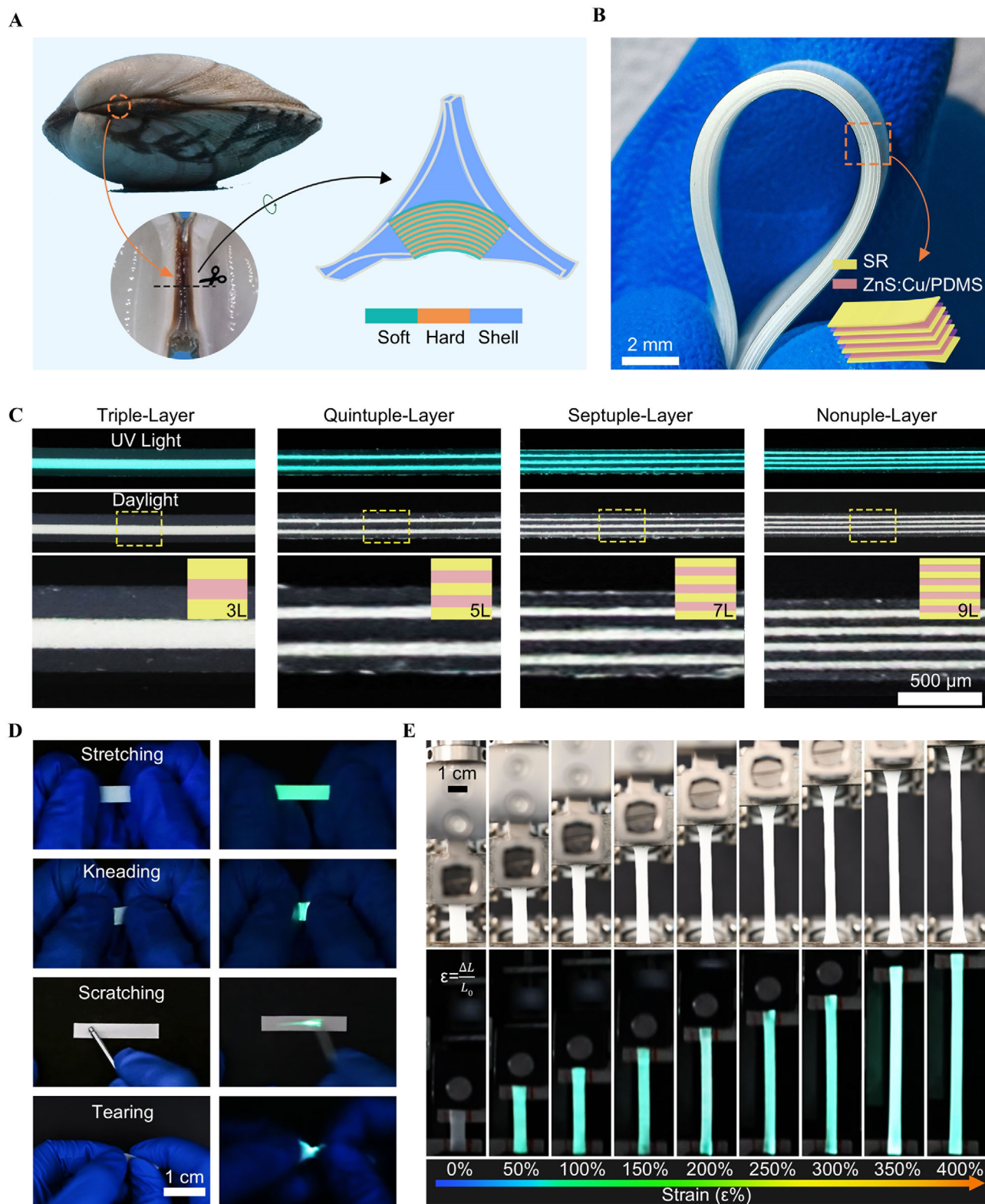


FIGURE 1 | Demonstration of the multilayer structure. (A) Photo of a clam, and schematic illustration of the shell at the cutting plane. (B) Photo of the representative 9L ML film. (C) Optical photos of the cross-sections of the 3L, 5L, 7L, and 9L ML films under 365 nm UV and daylight. (D) Photos of the 9L ML film under various mechanical stimuli. (E) Photos of the 9L ML film stretched from 0% to 400%.

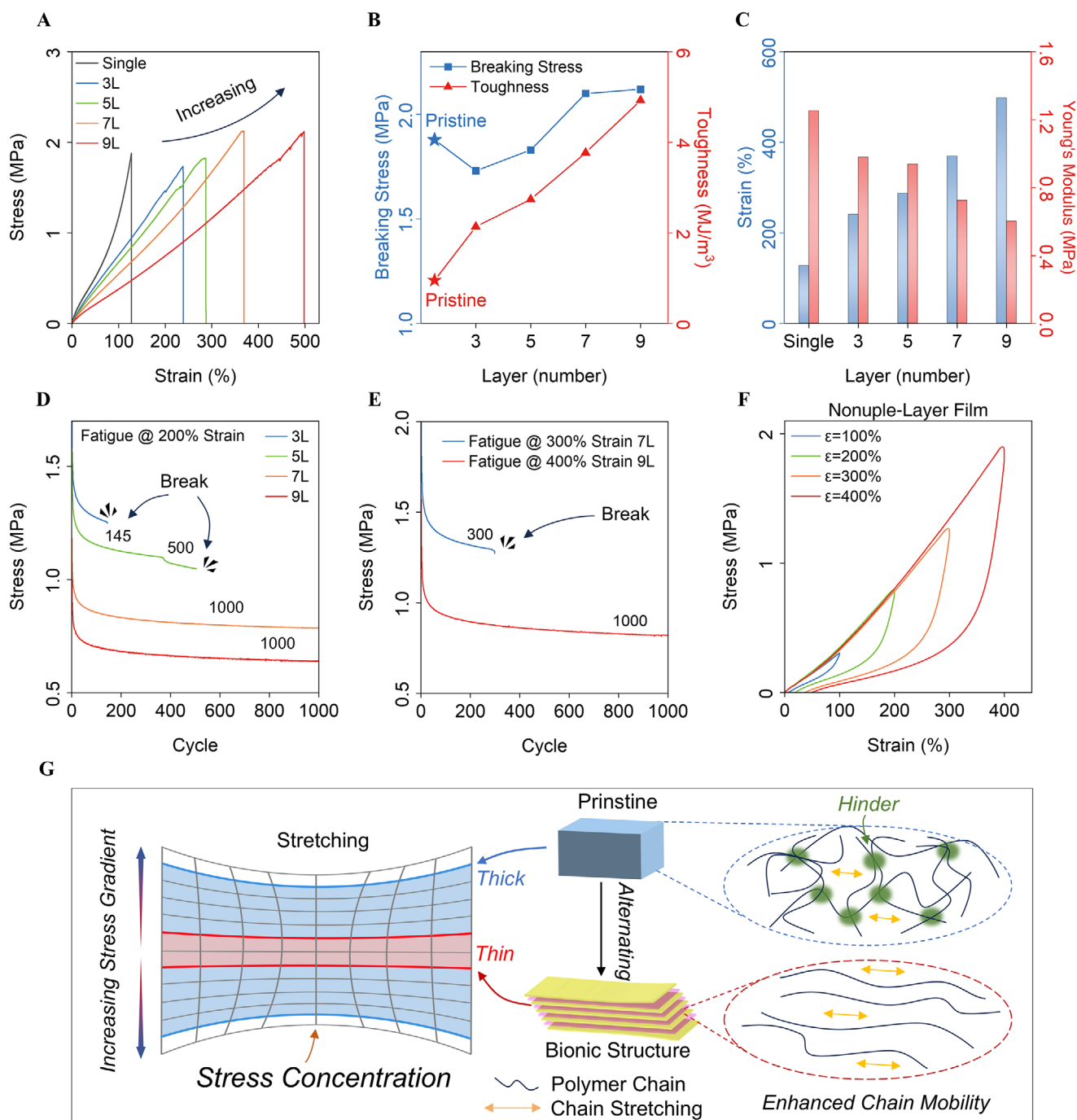


FIGURE 2 | Mechanical performance of multilayer film. (A) Stress–strain curves of the single-layer, 3L, 5L, 7L, and 9L ML films. (B) Breaking stress and toughness of the single-layer (pristine), 3L, 5L, 7L, and 9L ML films. (C) Ultimate tensile strain and Young's modulus of the single-layer and 3L–9L ML films. (D) Stress–Number of cycles (S–N) curves of the 3L, 5L, 7L, and 9L ML films at 200% tensile strain obtained from tensile fatigue tests. (E) S–N curves of the 7 and 9L ML films at ultimate strain obtained from tensile fatigue tests. (F) Stress–Strain curves of the 9L ML film at 100%–400% strain. (G) The mechanism for enhanced ductility of the multilayer structure.

contrast, as illustrated in Figure 2B, the ultimate fracture stress rises only modestly, from 1.88 to 2.12 MPa, while the slope of the stress–strain curves, i.e., elastic stiffness, diminishes sharply. These signatures confirm that the pronounced stretchability originates from the progressive softening of the ML strips toward the intrinsic compliance of the SR matrix. Accordingly, the fracture toughness, quantified as the area beneath the stress–strain curve, surges from 0.95 to 4.93 MJ m^{−3}. The scissor-like

inverse correlation between strain and Young's modulus in Figure 2C visually captures this transition, evidencing a seamless mechanical evolution from PDMS-like rigidity to SR-like elasticity. In contrast, the ML strip obtained by directly mixing PDMS and SR shows no significant increase in stretchability and exhibits a marked decrease in ML intensity (Figure S3), indicating that the enhanced performance arises from the multilayered architecture.

As the stretchability increases, the fatigue life of the ML strips is dramatically extended. Within the ultimate strain window of the pristine ZnS:Cu@Al₂O₃/PDMS ML strip, the engineered strips accumulate an excessive number of cycles, rendering lifetime assessment impracticable. Fatigue tests were therefore conducted at 200% strain (Figure 2D), revealing an exponential rise in cycle count with layer number: 145 cycles for the 3L sample, 500 for the 5L one, and >1000 for both 7 and 9L strips. Pushing further, the 7L ML strip endures 300 cycles at 300% strain, whereas the 9L one surpasses 1000 cycles at 400% strain (Figure 2E). Figure 2F presents the cyclic loading-unloading behavior of the 9L film at progressively higher strains from 100% to 400%. Residual stress rises only marginally and remains consistently low, underscoring the near-elastic response of the ML strip. Notably, the elastic modulus declines only during the first loading cycle, as a result of initial chain disentanglement, molecular orientation, and interfacial stress relaxation. Rapid stabilization is subsequently achieved, and no appreciable degradation or fluctuation is detected in subsequent cycles (Figure S4), indicating excellent structural stability and fatigue resistance under cyclic large-strain deformation. Even at the ultimate strain, creep is undetectable, suggesting an exceptional potential for high-precision stress sensing.

Figure 2G elucidates the mechanism behind the enhanced tensile ductility in multilayer structures. The layer thickness of composite elastomers directly determines stress transfer efficiency and fracture behavior [56]. In thick-layer ZnS:Cu@Al₂O₃/PDMS composites, the interior exhibits relatively lower stress, whereas the surface accumulates higher stress. This nonuniform distribution causes surface stress concentrations to reach the fracture strength ahead of the interior, triggering microcrack initiation at the surface and driving continuous crack propagation inward until catastrophic failure occurs. In contrast, thin-layer composites feature shortened stress transfer paths, enabling rapid and uniform stress distribution through the film. The diminished stress gradient ensures uniform deformation without localized stress overload, thereby suppressing premature fracture. Furthermore, the multilayer design reduces the constraints on molecular chains along the stretching direction by minimizing the thickness of individual layers. A thick single-layer film exhibits high molecular chain entanglement density, which impedes chain elongation [57]. Conversely, the multilayer architecture provides enhanced chain mobility for molecular chains during deformation, promoting more efficient chain elongation and ultimately enhancing the elongation at break.

2.3 | Strengthening and Toughening Mechanisms

As key metrics of mechanical degradation, crack initiation and growth in pristine and laminated ML strips were systematically examined. Figure 3A presents the cross-sectional SEM micrographs of the as-prepared specimens after fracture at ultimate strain and following 10 000 fatigue cycles (100% of strain for the pristine one, and 400% of strain for the 9L sample). The pristine PDMS strip exhibits a flat and featureless fracture surface, providing evidence of instantaneous brittle failure. This morphology reveals that the plasticity of PDMS-chain affords negligible energy dissipation, so virtually all released mechanical energy is funneled directly into rapid crack advance. In stark

contrast, the laminated strips display conspicuous staggered interlayer delamination. Each soft, high-toughness SR interlayer acts as an energy sink, blunting crack tips and dissipating fracture energy through extensive shear deformation and interfacial sliding, thereby arresting catastrophic failure and imparting pronounced toughness. Moreover, after 10 000 fatigue cycles, the monolithic film is riddled with pervasive microcracks, whereas the laminated strip exhibits only sparsely distributed submicron cavities. This striking contrast underscores the laminate's exceptional ability to suppress crack initiation and arrest early-stage damage, preserving structural integrity under prolonged cyclic loading.

Unilateral-notch testing was subsequently employed to quantify the crack-propagation resistance of the laminated strips. A 2-mm-diameter semicircular notch was introduced along one edge (Figure 3B). Even when stretched to 400%, the laminate remains intact and the notch shows minimal extension, evidencing pronounced crack-arrest capability. In stark contrast, the ultimate strain of the pristine strip plummets from 130% to 40% in the presence of the same defect (Figure S5). This behavior mirrors the fracture-morphology observations and corroborates that hierarchical lamination effectively suppresses crack advance.

To provide a more intuitive understanding of the reasons for the enhanced toughness of the multilayer film and its crack propagation resistance, finite element analysis (FEA) of the fracture process was performed. As shown in Figure 3C and Figure S6, 2 mm notches were introduced at four critical locations, i.e., the PDMS side (single-layer film), the SR side (single-layer film), the PDMS layer within the 9L structure, and the PDMS-SR interface. Significant stress concentration is observed in PDMS under strain loading (Figure S6A), resulting in rapid crack propagation and premature fracture at low strain levels. Conversely, the SR exhibits no observable crack extension under identical deformation conditions (Figure S6B). In the 9L composite film, while cracks are generated and extended within the PDMS layer, the high-toughness SR layer effectively inhibits further crack advancement (Figure S6C), thereby preventing catastrophic failure. Notably, when interfacial cracks are intentionally introduced at the PDMS-SR boundary (Figure 3C), the crack propagation velocity in the SR layer is markedly slower than that in PDMS under applied stress, confirming the inherent fracture resistance of SR. To further quantify the toughening effect, trouser tear tests, energy release rate (*G*) measurements, and single-edge notched tensile tests were performed (Figure S7, Figure 2A and Figure S8), all of which consistently demonstrated a substantial enhancement in fracture resistance (details are presented in Supporting Information). This synergistic design effectively suppresses both crack initiation and growth, enabling exceptional mechanical stability under extreme deformation.

2.4 | ML Performance

The ML performance of the strips was carefully characterized. As shown in Figure S9, the ML spectra of the single-layer and 9L strips under 100% strain exhibit identical spectral profiles and comparable emission intensities. This indicates that the multilayer architecture does not alter the luminescent properties, such

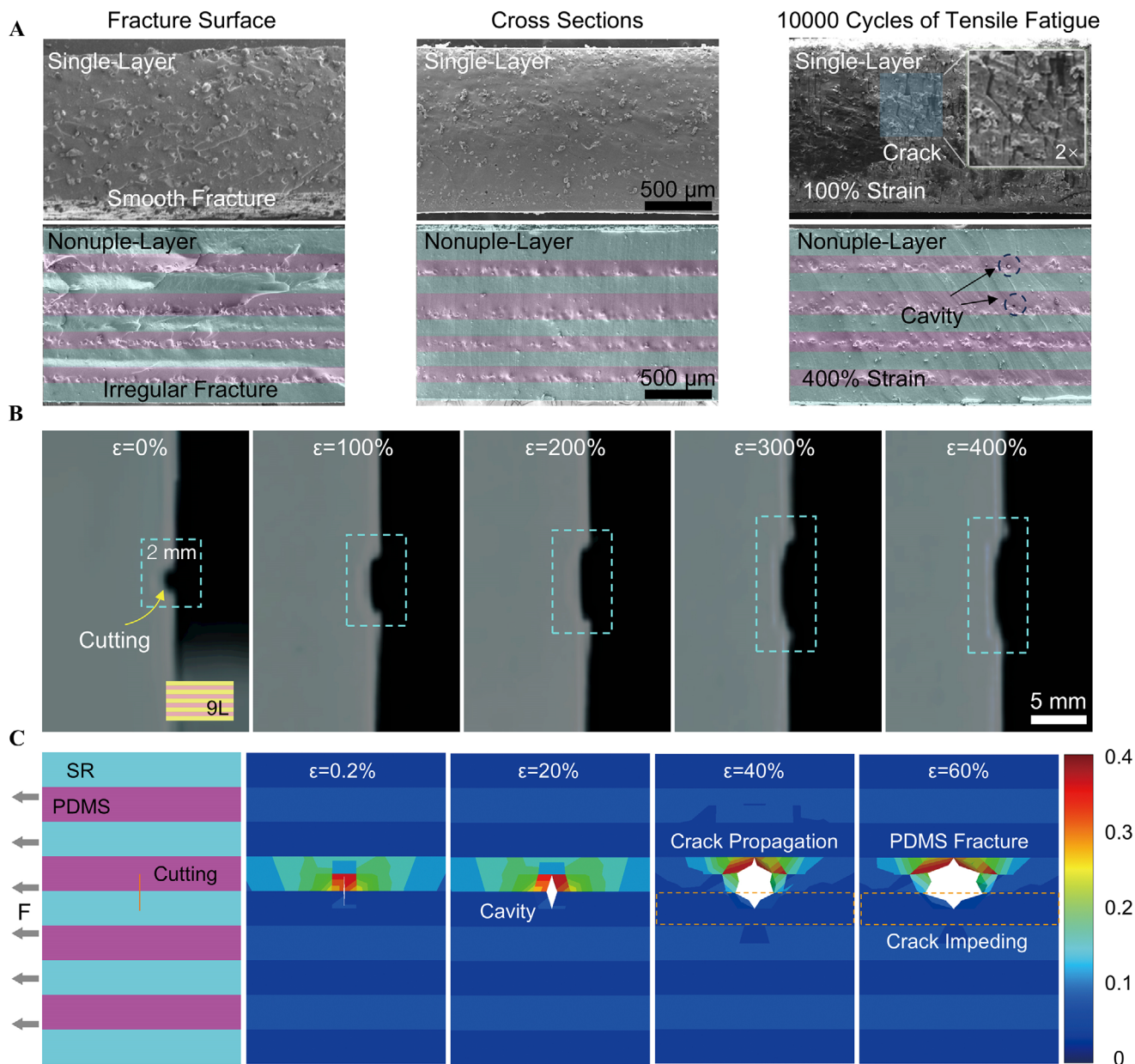


FIGURE 3 | Strengthening and toughening mechanisms. (A) Cross-sectional SEM micrographs showing single-layer and nonuple-layer samples in fractured (left), pristine (center), and ultimate tensile fatigue (right) conditions. (B) 9L ML film with unilateral notch at different strains. (C) Stress distribution plots of the notched 9L film under different strain loads calculated by FEA.

as the color or brightness. Therefore, the observed enhancement in ML performance is attributed to the improved mechanical characteristics afforded by the multilayer design, rather than to any direct modulation of the luminescence mechanism. Figure 4A illustrates the experimental configuration employed for ML measurements, and the ML intensity is extracted by first converting the captured ML images to greyscale and then integrating the grey levels over the entire emissive area. We first compared the ML brightness of strips with different layer counts under identical applied strain (Figure 4B). Strikingly, all samples exhibit almost identical ML intensities. This observation indicates that the ML brightness of different strips is dictated by strain rather than stress. According to previous literature, the driving energy for the $\text{ZnS:Cu@Al}_2\text{O}_3/\text{PDMS}$ composite originates from frictional interactions between the $\text{ZnS:Cu@Al}_2\text{O}_3$ phosphor

particles and the PDMS elastomer [58–60]. The absence of any significant influence from either impact energy or tensile rate on the ML of the strips, as shown in Figures S10 and S11, further supports this conclusion. Consequently, the strain-dependent nature of the ML intensity can be ascribed to the consistent relative displacement between the phosphor particles and the PDMS matrix across different strips when the same strain is applied. Notably, the multilayer strips become progressively softer as the layer number increases, a trend reflected in the monotonically decreasing Young's modulus (Figure 2C). Consequently, under an identical applied stress, the multilayer structure sustains a substantially larger strain, which translates into a higher stress sensitivity for the 9L strips. Figure 4C compares the correlation between ML intensity and strain (0%–100%) for single-layer and 9L films. A detectable ML signal emerges in the 9L film at a

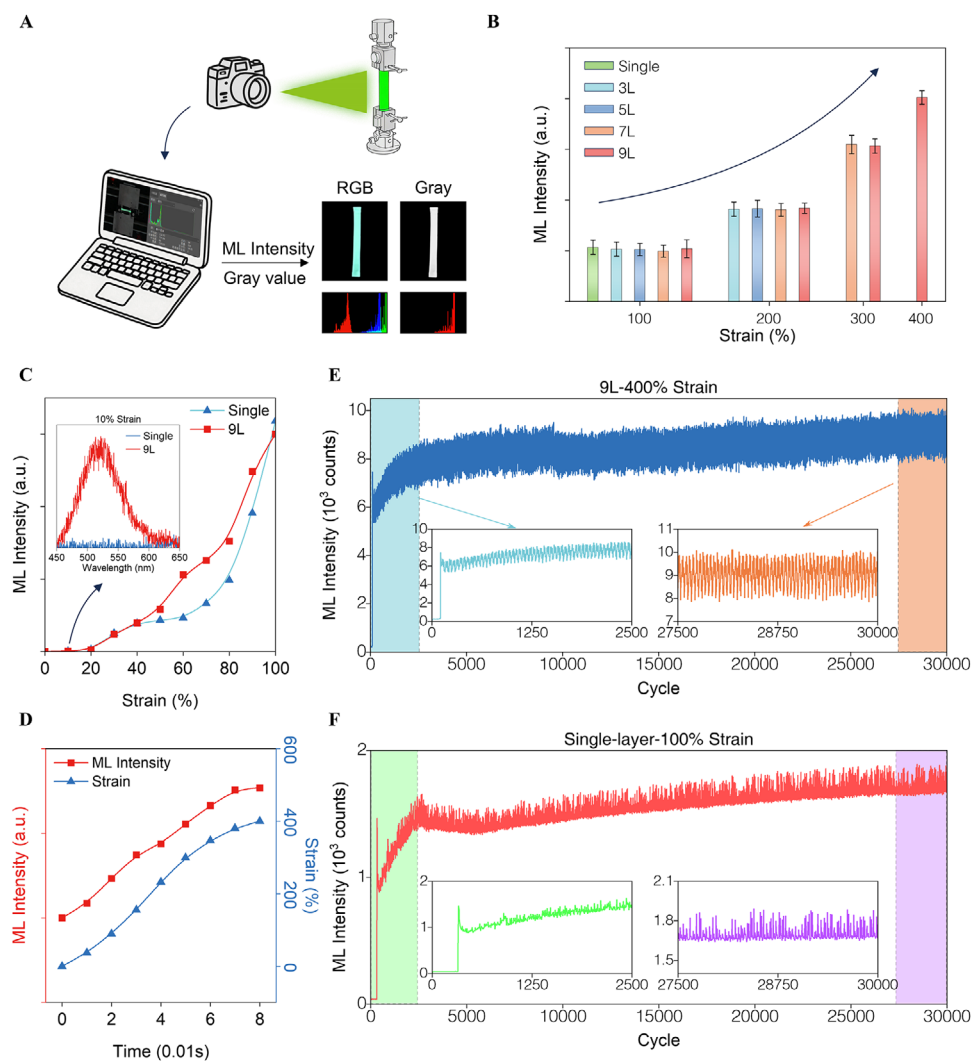


FIGURE 4 | ML performance of multilayer film. (A) Schematic of the tensile test setup. (B) ML intensity of the single-layer and the 3L–9L films under various tensile strains. (C) ML intensity of 9L and single-layer film changed with strain during tensile tests (strain: 0%–100%, strain rate: 200 mm s⁻¹). (D) ML intensity of 9L film changed with strain during tensile tests via image processing (strain: 0%–400%, strain rate: 200 mm s⁻¹). (E) The repeatability and cyclic stability of the 9L composite film exposed to a repeating mechanical stretching test. The repeatability tests were performed under a strain of 400%. (F) The repeatability and cyclic stability of single-layer film exposed to a repeating mechanical stretching test. The repeatability tests were performed under a strain of 100%.

strain as low as 10%, whereas no signal is detected in the single-layer film under the same condition. This reveals a reduced ML threshold in the 9L strip. We can therefore unequivocally conclude that the limited stretchability of ZnS:Cu@Al₂O₃/PDMS composites constitutes the primary constraint on their ultimate ML brightness. The bio-inspired multilayer architecture dramatically extends stretchability, widening the effective measurement range and elevating the ultimate brightness ceiling by more than four-fold. These metrics establish the 9L film as a record-performance PDMS-based flexible ML composite, as summarized in Table S1.

To elucidate the dynamic response of ML to strain, we synchronously recorded the ML intensity and applied tensile strain of the 9L film during a single rapid stretch from 0% to 400%. As illustrated in Figure 4D and Figure S12, the ML intensity rises monotonically and quasi-linearly with increasing strain, delivering a wide sensing range (0%–400%), underscoring the strip's

outstanding capability for real-time strain tracking/monitoring. Figure 4E,F present the ML durability of the 9L strip (at 400% of strain) and single-layer strip (at 100% of strain) under continuous cyclic stretching spanning the first 30 000 cycles, respectively. An obvious intensity spike is observed in the initial loading–unloading cycles (Figure 4E,F), which corresponds to the progressive separation of phosphor particles from the PDMS matrix. The detailed mechanism is discussed in the Supporting Information (Note S1). A bright ML signal is sustained as long as the film remains intact. Over this interval, the ML intensity rises initially and then plateaus at a saturation level, whose behavior is fully consistent with the triboelectric charge evolution trend (Figure S13). Because ZnS@Al₂O₃/PDMS is a prototypical tribo-ML system, the progressive accumulation of the tribo-charges with repeated deformation eventually levels off once a saturation charge density is reached, a phenomenon routinely observed in many triboelectric nanogenerator platforms [61–63].

2.5 | Mechanics Visualization for Intelligent Wearable Device

Based on the visualization ability, the multilayer composite film can be employed for high-level anticounterfeiting, self-powered displaying, and visualized mechanics sensing (as depicted in Figure S14), bringing effective solutions to the long-standing bottleneck issues in various fields. The multilayer structural films can be easily fabricated into arbitrary patterns. Without an external power source, these films display bright patterns upon mechanical stretching under dark conditions (Figure S14A). This deformation-induced pattern display capability provides new insights for advancing high-security anticounterfeiting technologies [64]. When writing on the multilayer ML film, the written patterns can be visually presented with distinct intensity distribution because of the significantly enhanced ML performance (Figure S14B). The spatially resolved luminescent patterning demonstrates the capability of the ML film to visualize mechanics in real time, which provides a new approach for mechanics analysis toward structural health in engineering. Furthermore, the multilayer film can distinguish among different types of mechanical stimuli. Under stretching, kneading, scratching, or tearing forces, the generated ML signals display distinct patterns (Figure S14C). Systematic monitoring of these patterns allows for accurate identification of the stimulus type. These ML signal characteristics enable the film to distinguish between mechanical stimuli, serving as a novel way for stimulus recognition in intelligent sensing systems.

In addition to the outstanding ML performance, the developed multilayer film also shows superior stretchability and toughness with remarkable anticrack activity. These make the devices fabricated from the multilayer film more comfortable to wear and more durable for use. In other words, the as-fabricated multilayer ML films are very suitable for intelligent wearable device applications. Herein, a flexible and wearable device was constructed as a representative utilizing the easy processability into fabric threads, as shown in Figure 5A. These fabric threads still have excellent resistance to tensile fracture and crack expansion. Even if many notches were artificially introduced in the fabric thread, the fabric thread is still able to withstand large deformation stretches. These fabric threads exhibit exceptional plasticity, enabling the fabrication of fabrics with diverse geometries (Figure S15). Based on these fabric threads, the smart ML fabric is obtained by knitting method. Obviously, the fabrics can show bright ML under stretching (Movie S3). This wearable and self-powered displaying device endows the clothes with more functionalities for information expression. Moreover, the emitted ML signal can also reflect the health status of the human via intensity sensing/monitoring. To tentatively prove this idea, an intelligent monitoring system was constructed based on the ML fabric (Figure 5B), composed of the ML fabric sensor, a photoelectric transducer, and a smartphone-integrated software platform. When mechanics is generated by human joint bending, ML emission is triggered through strain-to-light conversion. The emitted luminescent signal is acquired in situ and wirelessly by a commercial photoelectric sensor, where optical signals are transduced into electrical signals. These signals are transmitted wirelessly to a smartphone terminal via an analog-to-digital conversion module and analyzed algorithmically to reconstruct

mechanical information. Within the software interface, the historical maximum ML intensity is recorded in the upper panel, enabling retrospective assessment of joint health status (classified as safe or dangerous). Real-time ML signals are displayed in the lower panel for instantaneous motion state monitoring. Through the integration of preset critical thresholds and ML intensity-bending degree curve (Figure S16), the system is able to make timely judgments about the health status of the human during movements, thus reducing the further damage to the body.

Moreover, the as-constructed 9L film concurrently performs real-time self-diagnosis of its own structural health (Figure 5C). For example, a 2 mm diameter hole was artificially introduced into the samples, serving as a controlled defect to enable dynamic tracking of stress/strain distribution under loading conditions. Along with the stretching deformation of the sample, green light emission starts around the hole, corresponding to the initial stress concentration. With the progress of the stretching process, the luminescence propagation over a large region along the extensional direction is observed, making the stress/strain distribution become uniform. The above dynamic images allow us to visually observe the stress/strain heterogeneity induced by system defects, based on which the structural health of the system itself could be monitored.

3 | Conclusion

A bio-inspired high-low toughness ML multilayer structure was developed. Compared to the traditional PDMS-based ML film, the modified film exhibits a dramatic enhancement in stretchability, with fracture strain increasing from approximately 100% to 500%. This improvement is accompanied by more than a four-fold boost in ultimate ML intensity, along with sustained durability demonstrated over 30 000 cycles at the tensile limit. The reduced layer thickness and suppressed crack propagation by the high-toughness SR interlayer raise toughness from 0.95 to 4.93 MJ m⁻³. The toughening mechanism of the multilayer ML composite film is attributed to the reduced spatial confinement effect of molecular chains resulting from the diminished thickness of individual layers within the multilayer architecture, coupled with the inhibitory influence of the high-toughness SR layer on microcrack propagation within the PDMS. Based on the desirable overall performance of the ML composites, a wearable human motion monitoring system was developed for remote and real-time health monitoring. This work breaks through the performance limit of PDMS-based flexible ML films and develops flexible force-emitting films that combine high mechanical properties with high ultimate brightness, paving the way for the development of distributed intelligent stress/strain sensors and promoting the practical application of flexible ML films.

4 | Experimental Section

4.1 | Materials

Copper-doped zinc sulfide (ZnS:Cu@Al₂O₃) was purchased from Shanghai Keyan Phosphor Technology Co., Ltd. PDMS of Sylgard184 was purchased from Dow Corning, and silicone rubber

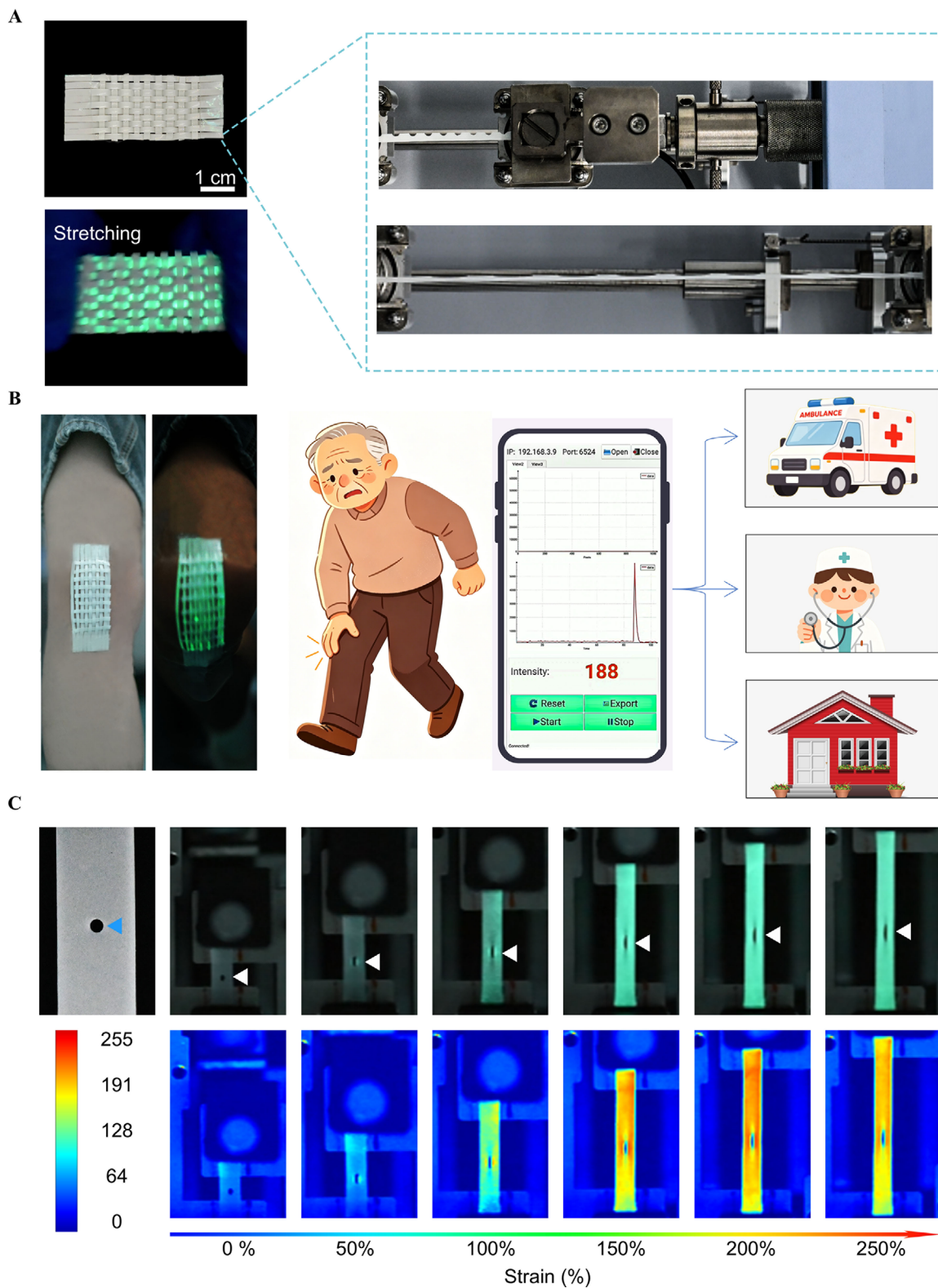


FIGURE 5 | The application of multilayer film in stress visualization and intelligent wearable devices. (A) Photos of smart fabric by knitting method and ML effects. (B) Photos and schematic diagram of wireless human movement status monitoring system. (C) ML photos and mappings to provide the spatial defect evolution in the 9L ML film for self-diagnosis.

(SR) was purchased from Shenzhen Guoyuan Technology. They were used as received without further purification.

4.2 | Fabrication of ZnS:Cu@SR_x•PDMS_{1-x} Composite Film

First, PDMS base resin (4.0 g) and curing agent (0.4 g) in a mass ratio of 10:1 were mixed in a petri dish (diameter: 30 mm). Then, 2.0 g of ZnS:Cu@Al₂O₃ powders were uniformly dispersed into the aforementioned PDMS precursor through mechanical stirring for 5 min to obtain Mixture A (ZnS:Cu@Al₂O₃/PDMS). ZnS:Cu@Al₂O₃/SR composite material was prepared following a similar procedure to that of ZnS:Cu@Al₂O₃/PDMS, except that the ratio of base resin, curing agent, and powder for ZnS:Cu@Al₂O₃/SR was set to 1:1:1, yielding Mixture B. Mixtures A and B were then physically blended in mass ratios of 1:0, 0.8:0.2, 0.6:0.4, 0.4:0.6, 0.2:0.8, and 0:1. Subsequently, the blended mixtures were poured into dumbbell-shaped PTFE molds (25 mm × 10 mm × 2 mm), and degassed under negative pressure (−80 kPa) for 30 min. After curing at 70°C for 0.5 h, standard ZnS:Cu@SR_x•PDMS_{1-x} composite film test samples were obtained.

4.3 | Fabrication of Multilayer Structure ML Films

The process flow diagram (Figure S17) illustrates the entire fabrication procedure of the bioinspired multilayer mechano-luminescent films, including precursor preparation, alternating spin-coating, step-by-step heat treatment, and final heat treatment. PDMS base resin (2.0 g), ZnS:Cu@Al₂O₃ (1.0 g) powders and curing agent (0.2 g) in the mass ratio of 10:5:1 were homogeneously mixed in a Petri dish (diameter: 30 mm) to obtain Mixture C. The silicone rubber (2.0 g) and curing agent (2.0 g) were mixed homogeneously (5 min) in a petri dish (diameter: 30 mm) in a 1:1 mass ratio to obtain Mixture D. Employing the spin-coating technique, the petri dish (diameter: 30 mm) was evenly coated with Mixture D, placed in the oven at 50°C for 5 min, and then evenly coated with Mixture C. Repeating this process several times to obtain a multilayer structured films. By manipulating the number of coating cycles, 3-layer (3L), 5-layer (5L), 7-layer (7L), and 9-layer (9L) films were prepared. After curing at 70°C for 0.5 h, standard testing samples of ZnS:Cu@Al₂O₃/PDMS-SR multilayer structure ML films were obtained.

4.4 | Surface Triboelectric Potential Measurement

The surface triboelectric potential was measured using a commercial electrostatic measurement probe (SK050, KEYENCE, Japan) featuring high input impedance and high sampling rate, coupled to a self-built tensile apparatus. The sample was tailored into a dumbbell-shaped specimen with a gauge length of 20 mm and a width of 10 mm, and the total thickness was fixed at 1 mm. The measurement probe was positioned 5 mm above the central surface to maintain a constant detection distance. Cyclic stretching was performed at 400% strain with a stretching rate of 100 mm s^{−1}, and the potential signals were monitored in real time over 30 000 cycles.

4.5 | Fracture Mechanics Characterizations

Fracture mechanics characterizations were performed on a universal tensile testing machine (Shimadzu AGS-X-10 kN) at 50 mm min^{−1}. For trouser tear tests, samples (40 mm × 20 mm × 1 mm) with an initial notch length of 20 mm were tested with pneumatic fixtures to prevent sliding. For energy release rate (*G*) measurements, unnotched specimens were tested with a clamping height of 20 mm. For critical fracture energy (*G_c*) measurements, single-edge notched samples (40 mm × 20 mm × 1 mm) with a 5 mm notch were tested, and *G_c* was calculated using the Green–Smith method. All elastomer samples underwent three replicate tests.

4.6 | Characterizations

The morphology information was obtained by the SEM (Tescan Clara GHM). The mechanical properties were evaluated using a universal tensile testing machine (Shimadzu AGS-X-10 kN) at a strain rate of 100 mm min^{−1} at room temperature. The composite elastomers were stretched on a homemade tensile apparatus, and the ML spectra were in situ collected and recorded by a fluorescence spectrophotometer (Omni-λ300i, Zolix Instruments Co., Ltd.) linked to a charge-coupled device (CCD) camera (iVac-316, Edmund Optics Ltd.). The ML intensity is extracted by converting the acquired ML images to grayscale and subsequently integrating the grayscale values across the entire emission area. The stress state is simulated by finite element analysis (FEA) software (SALOME-MECA). Additionally, the optical images were captured using a digital camera (SONY ILCE-7M4).

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File 1: adma74267-sup-0001-SuppMat.docx.

Supporting File 2: adma74267-sup-0002-MovieS1.mp4.

Supporting File 3: adma74267-sup-0003-MovieS2.mp4.

Supporting File 4: adma74267-sup-0004-MovieS3.mp4.